jal-netra

Jal-Netra: manufacturing and build manual (prototype programme)

> **Status: design, simulation and a rule-checked board. Nothing has been built or measured on water.** The product is **not manufacturable for sale today**: the acoustic signal has never been measured (Gate 1 can sink the architecture), production materials for drinking-water contact are not selected, and the economics only work above about Rs 7,000. This manual describes how to build and test **prototype units** and what must be passed before any production step. Follow the order: **mechanics and Gate 1 first, board second.**

1. Build order (do not skip)

1. **Print/make one spool** (section 3) and run **Gate 1** with a signal generator and oscilloscope **before spending money on boards**. If the received amplitude at the piezo is below 0.3 mV, redesign (more drive voltage, matching layer, shorter path, 2 MHz) rather than build more.
2. Order the PCB (section 4) and assemble 2-3 units.
3. Flash firmware, run Gates 2-5 (`docs/TEST_PLAN.md`).
4. Only after the gates: material selection, certification, cost-down, production plan.

2. What is in the package

Item	Where	State
Main PCB (86.1 x 66.1 mm, 2 layers, 78 parts)	`jal-netra/hardware/` (`1_fabrication` ... `5_verification`)	all automatic checks pass: DRC 0, netlist match, verified LCSC numbers; **layout is software-generated and unreviewed**
Spool CAD (Z-body 99 x 64 mm, 20 ml fluid volume, 4 mm walls)	`jal-netra/cad/spool_body.stl`, `spool_fluid_volume.stl`, `piezo_disc_reference.stl`, `spool_info.json`	watertight; generated by `cad/spool.py`
Firmware core (tested) + STM32G4 skeleton	`jal-netra/firmware/lib/jn_core.c`, `firmware/src/main_stm32g4.c`	core: 18 host checks pass; skeleton **not built for the target**
Simulations	`jal-netra/sim/`	physics, digital twin, SPICE
Test plan	`jal-netra/docs/TEST_PLAN.md`	six gates

3. The spool (mechanics)

- Two 1 MHz PZT discs, 10 mm diameter, face each other across a 60 mm bore of 12 mm diameter; brass or polymer body; side ports (Z-body).
- **Prototype body:** print `spool_body.stl` in SLA resin or SLS (PA12) at 100 % infill, pressure-test to 10 bar. **Printed PA12 is not acceptable for drinking-water contact**; the prototype is for bench water only. Production needs brass (e.g. dezincification-resistant) or a certified polymer: not selected.
- Piezo mounting: bond each disc in its pocket with a thin, void-free epoxy layer (the water-facing electrode must not be shorted), potted behind; solder 2-core cables (tin 22 AWG) to the faces; check disc

capacitance and resonance with an LCR meter (1 MHz +-5 %); **select matched pairs** (> 3 dB mismatch breaks Gate 1).
- Remove all air (tilt and tap); bubbles kill the signal.
- Inlet/outlet: 1/2 in BSP fittings, PTFE tape; pressure test with water at 10 bar for 1 h (Gate 5) behind a shield.

4. The board

1. Order PCBs and SMT assembly per `manuals/STANDARD_PROCEDURES.md` A and B using `hardware/1_fabrication/jal_netra_gerbers.zip` and `hardware/2_assembly/`. The STM32G431CBT6, OPA356 (x2), TS5A3159 (x2), AP2112K-3.3, BAV99 (x2), ferrite, 8 MHz crystal (12 pF) and the 0603 passives are on the JLC BOM. Through-hole/hand parts: USB-C shell tabs, JST-PH connectors (transducers A and B, OLED, UART), SWD header, buzzer, test button, test points.
2. **Review before fab (recommended):** a PCB engineer should review the receive-chain layout (guarding, star ground, TX bridge return paths). Production should be 4-layer.
3. Solder the hand parts. Check no short between +3V3, AVDD and GND before power.
4. Power from a 5 V USB source with a current limit of 100 mA: +3V3 = 3.3 V, AVDD = 3.28 V, VMID = 1.64 V (README bring-up).

5. Programming

SWD with an ST-Link V2: `st-flash write jal_netra.bin 0x08000000` (or OpenOCD with `interface/stlink.cfg`, `target/stm32g4x.cfg`). The firmware around `jn_core` is a skeleton: the TIM3 burst generation, ADC1 4 MS/s capture, mux control and OLED output must be completed and verified on the bench; budget weeks for that. **No production firmware exists.**

6. Calibration and tests

Run Gates 1-5 exactly as written in `docs/TEST_PLAN.md`: acoustic signal (>= 1 mV target), zero stability (< 100 ps rms after fit), gravimetric flow calibration (< 3 % of reading, 0.2-20 L/min at 15 and 35 C), 14-day leak test, environment/abuse. Record each gate on `manuals/templates/eol_test_sheet.md` (add the gate rows).

7. Before any sale

Drinking-water contact material approval, legal-metrology position (it is a monitor, not a billing meter: do not claim billing accuracy), electrical safety of the USB supply, EMC, e-waste EPR, a cost-down (BOM about Rs 2,970; contribution about zero at Rs 5,499: price needs about Rs 7,000+ or a cheaper amplifier/transducer).

8. Known limits

No acoustic measurement exists; systematics (crosstalk, echoes, mismatch, bubbles, scale, thermal gradients, zero drift) are untested; profile factor and loss coefficient are textbook assumptions; accuracy targets are targets.